

Your codebase forms a **Spring Boot application** that enables real-time fraud detection using an **XGBoost model** exposed via a Python Flask API. Here's a concise breakdown of each file's role in retrieving and displaying predictions:

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#### **FraudDetectionService.java**

- **Purpose:** Connects to the Python-based ML API.
  - **Key Function:** `checkFraud()`:
    - Sends a POST request to `http://localhost:5000/predict` with the transaction data.
    - Returns the **raw JSON string** containing the XGBoost model's prediction.
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#### **FraudDetectionController.java**

- **Purpose:** Acts as an API controller to push prediction data to the frontend.
  - **Key Endpoint:** POST `/api/update`
    - Receives prediction data (e.g., `{transaction, prediction}`).
    - Uses `SimpMessagingTemplate` to **broadcast** the data to all subscribers via `/topic/transactions`.
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#### **WebSocketConfig.java**

- **Purpose:** Configures WebSocket communication using STOMP.
  - **Key Configs:**
    - `/fraud-websocket`: STOMP endpoint used by the frontend to establish a connection.
    - `/topic`: Prefix for messages sent **from server to clients**.
    - `/app`: Prefix for messages sent **from clients to server** (not used here).
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#### **FraudDetectionApplication.java**

- **Purpose:** Main entry point for the Spring Boot app.
  - **Bean:** Registers a `RestTemplate` bean to be used for HTTP requests (i.e., in the service).
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#### **application.properties**

- **Purpose:** Provides configuration for:
  - MySQL DB (used for storing transaction data).
  - Thymeleaf templating.

- Logging and port configuration.
  - Enables WebSocket message debugging.
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### build.gradle

- **Purpose:** Specifies project dependencies.
    - Includes Spring Boot, WebSocket, JPA, MySQL, WEKA (ML library), and Lombok.
    - Includes :ml-library module, suggesting a custom Java-based ML component is integrated too.
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### index.html

- **Purpose:** Frontend interface to show real-time predictions.
  - **Functionality:**
    - Establishes a WebSocket connection to /fraud-websocket.
    - Subscribes to /topic/transactions.
    - Displays each prediction with:
      - Transaction details
      - Whether it's **fraud** or **safe**
      - Prediction **probability**
    - Alerts the user if fraud is detected.
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